**Boyle's law**

**Boyle's law** (sometimes referred to as the **Boyle–Mariotte law**, or **Mariotte's law**) is an experimental gas law that describes how the pressure of a gas tends to increase as the volume of the container decreases. A modern statement of Boyle's law is

The absolute pressure exerted by a given mass of an ideal gas is inversely proportional to the volume it occupies if the temperature and amount of gas remain unchanged.

Mathematically, Boyle's law can be stated as

$$P \alpha \frac{1}{V}$$

or

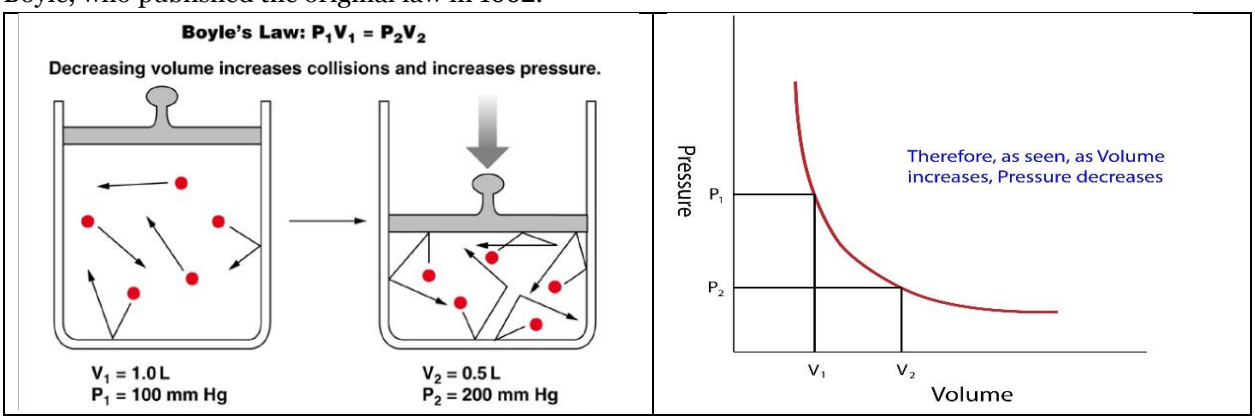
$$PV = k$$

where $P$ is the pressure of the gas, $V$ is the volume of the gas, and $k$ is a constant.

The equation states that product of pressure and volume is a constant for a given mass of confined gas as long as the temperature is constant. For comparing the same substance under two different sets of conditions, the law can be usefully expressed as

$$P_1V_1 = P_2V_2$$

The equation shows that, as volume increases, the pressure of the gas decreases in proportion. Similarly, as volume decreases, the pressure of the gas increases. The law was named after chemist and physicist Robert Boyle, who published the original law in 1662.




**Problem 1: A gas has volume 2L at atmospheric pressure. If the gas pressure changed to 2atm, find the volume of the gas.**

**Charles' law**

**Charle's law** (also known as the **law of volumes**) is an experimental gas law that describes how gasestend to expand when heated. A modern statement of Charles' law is:

When the pressure on a sample of a dry gas is held constant, the Kelvin temperature and the volume will be directly related.[1]

This directly proportional relationship can be written as:

$$V \alpha T$$

or

$$\frac{V}{T} = k$$

where:

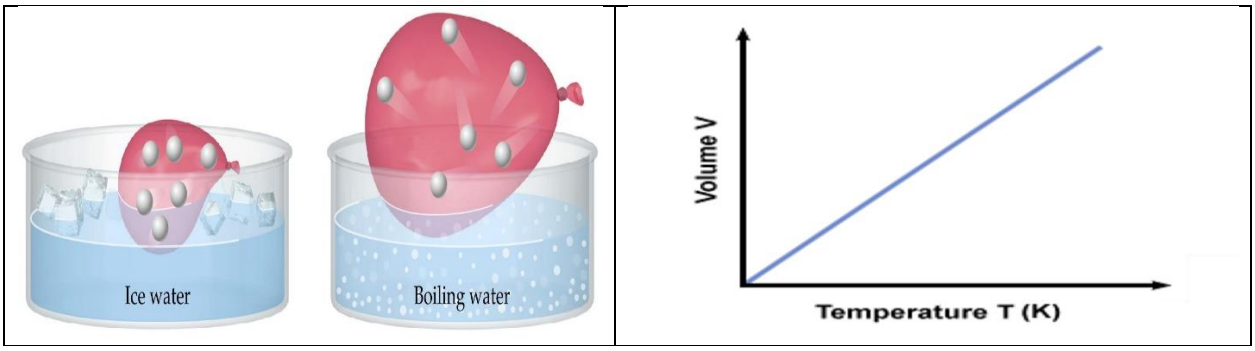
$V$ is the volume of the gas,

$T$ is the temperature of the gas (measured in kelvins),
$k$ is a constant.

This law describes how a gas expands as the temperature increases; conversely, a decrease in temperature will lead to a decrease in volume. For comparing the same substance under two different sets of conditions, the law can be written as:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{or} \quad \frac{V_2}{V_1} = \frac{T_2}{T_1} \quad \text{or} \quad V_1T_2 = V_2T_1$$

The equation shows that, as absolute temperature increases, the volume of the gas also increases in proportion.




**Gay-Lussac's law**

This law is often referred to as **Gay-Lussac's law of pressure–temperature,** between 1800 and 1802, discovered the relationship between the pressure and temperature of a fixed mass of gas kept at a constant volume. Gay Lussac discovered this while building an "air thermometer".

**The pressure of a gas of fixed mass and fixed volume,is directly proportional to the gas's absolute temperature.**

If a gas's temperature increases, then so does its pressure if the mass and volume of the gas are held constant. The law has a particularly simple mathematical form if the temperature is measured on an absolute scale, such as in kelvins. The law can then be expressed mathematically as

$$P \propto T$$

or

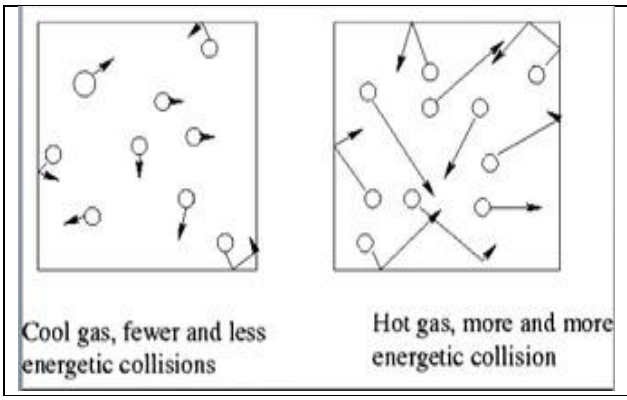
$$\frac{P}{T} = k$$

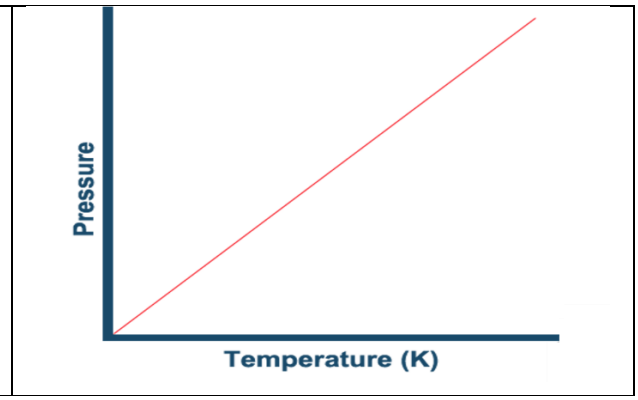
where:
$P$ is the pressure of the gas,
$T$ is the temperature of the gas (measured in kelvins),
$k$ is a constant.

This law holds true because temperature is a measure of the average kinetic energy of a substance; as the kinetic energy of a gas increases, its particles collide with the container walls more rapidly, thereby exerting increased pressure.

For comparing the same substance under two different sets of conditions, the law can be written as:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \text{or} \quad P_1T_2 = P_2T_1$$




**Avogadro's law**

Avogrado's law (sometimes referred to as **Avogadro's hypothesis** or **Avogadro's principle**) is an experimental gas law relating volume of a gas to the amount of substance of gas present. A modern statement of Avogadro's law is:

Avogadro's law states that, "equal volumes of all gases, at the same temperature and pressure, have the same number of molecules".

For a given mass of an ideal gas, the volume and amount (moles) of the gas are directly proportional if the temperature and pressure are constant.

which can be written as:

$$V \alpha n$$

or

$$\frac{V}{n} = k$$

where:

$V$ is the volume of the gas

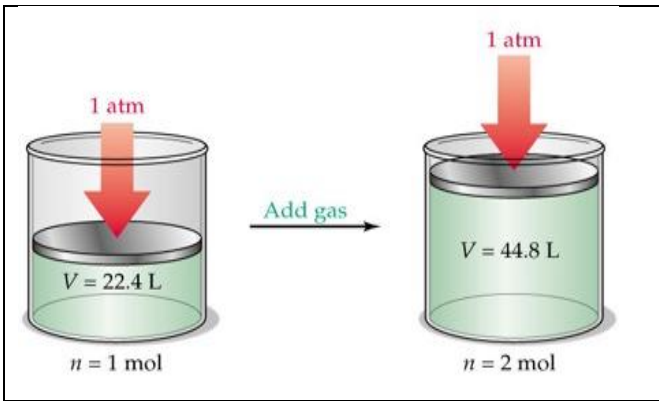
$n$ is the amount of substance of the gas (measured in moles).

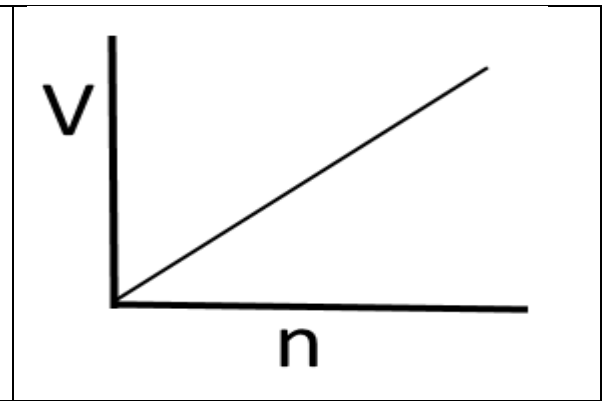
$k$ is a constant equal to RT/P, where R is the universal gas constant, T is the Kelvin temperature, and P is the pressure. As temperature and pressure are constant, RT/P is also constant and represented as $k$. This is derived from the ideal gas law.

This law describes how, under the same condition of temperature and pressure, equal volumes of all gases contain the same number of molecules. For comparing the same substance under two different sets of conditions, the law can be usefully expressed as follows:

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \quad \text{or} \quad V_1 n_2 = V_2 n_1$$

The equation shows that, as the number of moles of gas increases, the volume of the gas also increases in proportion. Similarly, if the number of moles of gas is decreased, then the volume also decreases. Thus, the number of molecules or atoms in a specific volume of ideal gas is independent of their size or the molar mass of the gas.






**What is an ideal gas?**

Gases are complicated. They're full of billions and billions of energetic gas molecules that can collide and possibly interact with each other. Since it's hard to exactly describe a real gas, people created the concept of an **Ideal gas** as an approximation that helps us model and predict the behavior of real gases. The term ideal gas refers to a hypothetical gas composed of molecules which follow a few rules:

1. **Ideal gas molecules do not attract or repel each other**. The only interaction between ideal gas molecules would be an elastic collision upon impact with each other or an elastic collision with the walls of the container.

2. **Ideal gas molecules themselves take up no volume**. The gas takes up volume since the molecules expand into a large region of space, but the Ideal gas molecules are approximated as point particles that have no volume in and of themselves.

If this sounds too ideal to be true, you're right. There are no gases that are exactly *ideal*, but there are plenty of gases that are close enough that the concept of an ideal gas is an extremely useful approximation for many situations. In fact, for temperatures near room temperature and pressures near atmospheric pressure, many of the gases we care about are very nearly ideal.

If the pressure of the gas is too large (e.g. hundreds of times larger than atmospheric pressure), or the temperature is too low (e.g. $-200°C$) there can be significant deviations from the ideal gas law.

**Molar form of the ideal gas law**

The pressure, $P$, volume $V$, and temperature $T$of an ideal gas are related by a simple formula called the **ideal gas law**. The simplicity of this relationship is a big reason why we typically treat gases as ideal, unless there is a good reason to do otherwise.

$$PV=nRT$$

Where $P$ is the pressure of the gas, $V$ is the volume taken up by the gas, $T$ is the temperature of the gas, $R$ is the gas constant, and $n$ is the number of moles of the gas.

Perhaps the most confusing thing about using the ideal gas law is making sure we use the right units when plugging in numbers. If you use the gas constant $R$=8.31$J/K\ mol$, then you must plug in the pressure $P$ in units pascals $Pa$, volume $V$ in units of $m^3$ and temperature $T$ in units kelvin.

**Molecular form of the ideal gas law**

If we want to use $N$ number of molecules, instead of $n$ moles, we can write the ideal gas law as,

$$PV = Nk_BT$$

Where $P$ is the pressure of the gas, $V$ is the volume taken up by the gas, $T$ is the temperature of the gas, $N$ is the number of molecules in the gas, and $k_B$ is Boltzmann's constant, $k_B = 1.38\times10^{-23} J/K$